STANDARD 8 – BIOLOGY

ANSWER KEY & SCORING SCHEME (ENGLISH VERSION)

Class: Standard VIII | **Subject: Basic Science (Biology)** | **Total Score: 20**

Question 1 — Score: 1

c) i, iii correct (Score: 1)

Question 2 — Score: 1

b) P-iii, Q-iv, R-i, S-ii (Score: 1)

Question 3 — Score: 2

This idea is completely appropriate.
It is a method of carrying out cattle rearing, poultry farming, and fish farming together complementarily along with plant cultivation.
The waste/by-products from one farming activity (e.g., cow dung, poultry manure) can be utilized as fertilizer and feed for another.
This significantly reduces farming expenses and helps ensure additional income and food security for the farmer.

Question 4 — Score: 2

- **A:** Prokaryotic cell / Bacterial cell (Score: ½)
- **B:** Eukaryotic cell (Score: ½)
- **C:** No distinct membrane-bound nucleus or membrane-bound cell organelles (Score: ½)
- **D:** Distinct membrane-bound nucleus and cell organelles are present (Score: ½)

Question 5 — Score: 2

5. (A)
a) The statement is completely true. By developing Genetically Modified (GM) crops with inbuilt pest resistance (e.g., Bt cotton) through genetic engineering, excessive use of chemical pesticides and its harmful environmental/health issues can be avoided (Score: 1).
b) Difference: Plantlets obtained through tissue culture have the exact same characteristics as the parent plant (no genetic change occurs). But in GM crops, desired foreign genes from other organisms are inserted to deliberately introduce new traits into the genetic structure (Score: 1).

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5. (B)
The statement is correct. The goal is to suppress pest populations below economic damage levels rather than completely eradicating all pests.
Chemical pesticide application is minimized by using biological and mechanical methods such as friendly insects, nets, traps, and pest-resistant seeds.
This helps maintain natural soil fertility, protects environmental balance, and ensures sustainable farming (Score: 2).

Question 6 **Score: 2**

6. (A)

Macronutrients: Elements required in large quantities by plants.

Example: Nitrogen (N), Phosphorus (P), Potassium (K), Calcium (Ca), Magnesium (Mg), Sulphur (S).

Micronutrients: Elements required only in very small (trace) quantities by plants.

Example: Iron (Fe), Zinc (Zn), Boron (B), Copper (Cu), Manganese (Mn), Molybdenum (Mo), Chlorine (Cl).

--- OR ---

6. (B)

Soil Testing: Helps accurately identify nutrient levels and pH value (acidity/alkalinity) in the soil. Thus, the exact required fertilizers and their precise dosage for the farmland can be determined.

Fertilizer Application: Provides balanced nutrients to plants and ensures healthy growth and high yield. It also avoids excessive fertilizer application and preserves soil health.

Question 7 **Score: 2**

The larger host cell engulfs an aerobic bacterium capable of aerobic respiration.

Instead of digesting the smaller cell, the larger host cell provides protection and shelter to it.

Subsequently, this mutualistic smaller cell permanently integrated and evolved over time into the **mitochondria** in eukaryotic cells.

Question 8 **Score: 2**

- **Krishi Bhavan:** Training programmes, technical advice, distribution of quality seeds/saplings.
- **Local Self-Government Institutions:** Financial subsidies, welfare schemes, awards, and recognitions.
- **Cooperative Banks** / **Societies:** Low-interest agricultural loans, procurement, and marketing assistance.
- **Online Apps** / **Platforms:** Solutions for pest infestations, weather warnings, daily market information.

Question 9 **Score: 3**

i) Conditions favoring the formation of simple organic molecules: (Score: 2)

- Gases in the primitive atmosphere (Methane, Ammonia, Hydrogen, Water vapor; absence of free oxygen).
- High temperature in the atmosphere.
- Energy sources: Continuous lightning, ultraviolet rays, solar energy, volcanic eruptions.
- Prolonged torrential rainfall caused by the condensation of water vapor and the formation of oceans.

ii) Miller-Urey Experimental Evidence: (Score: 1)

Yes. By artificially simulating the atmospheric conditions of primitive Earth in the laboratory, they proved that amino acids (organic molecules), the building blocks of proteins, could be synthesized from simple inorganic gases.

Question 10 **Score: 3**

10. (A)

a) Advantages of Nano-fertilizers: Being in nano-particle form, they can be absorbed very rapidly by plants. Required only in very small quantities; they do not alter soil structure unlike conventional synthetic fertilizers (Score: 1).

b) Limitations of Organic Fertilizers: Nutrient concentration is relatively low; it takes time for nutrients to be released through decomposition processes (Score: 1).

c) Role of Bio-fertilizers: Enriches natural soil fertility by supplying microorganisms (such as *Rhizobium*, *Azospirillum*, *Mycorrhiza*) that fix atmospheric nitrogen or solubilize phosphorus for easy uptake by plants (Score: 1).

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10. (B)

a) The statement is correct. In drip irrigation, water is delivered drop by drop directly to the roots through pipes, thereby avoiding evaporation loss. In wick irrigation, water is directly transported to plant roots using cotton wicks. Both methods help cultivate crops with minimum water loss (Score: 2).

b) Modern Farming Method: Vertical Farming / Pot culture / Terrace farming.

Another Benefit: Higher yield in limited space; year-round cultivation without being affected by climate fluctuations (Score: 1).